CARBONATURE · DATABASE & MRV AI-MODULE

Soil-carbon MRV — Data Flow & System Architecture

The verification backbone for Verra VM0042 v2.2 agricultural carbon projects: how field evidence becomes an audit-ready, credited number, and the seven schema domains that hold it.

Engine PostgreSQL 16 · PostGIS · pgvector **Schema** mrv · 41 tables · 4 views **Host** AWS RDS eu-west-1 **Status** 7/7 stages live

---

- Verification lineage
- Commercial & GHG accounting
- Spatial core
- AI / agent layer
- Governance & audit

FIG 1 Data Flow

Left to right: a physical soil sample and a farm's activity records enter, are validated and stored append-only, drive the SOC and GHG computation, and roll up into credits and Verra units — every step written to an immutable audit log.

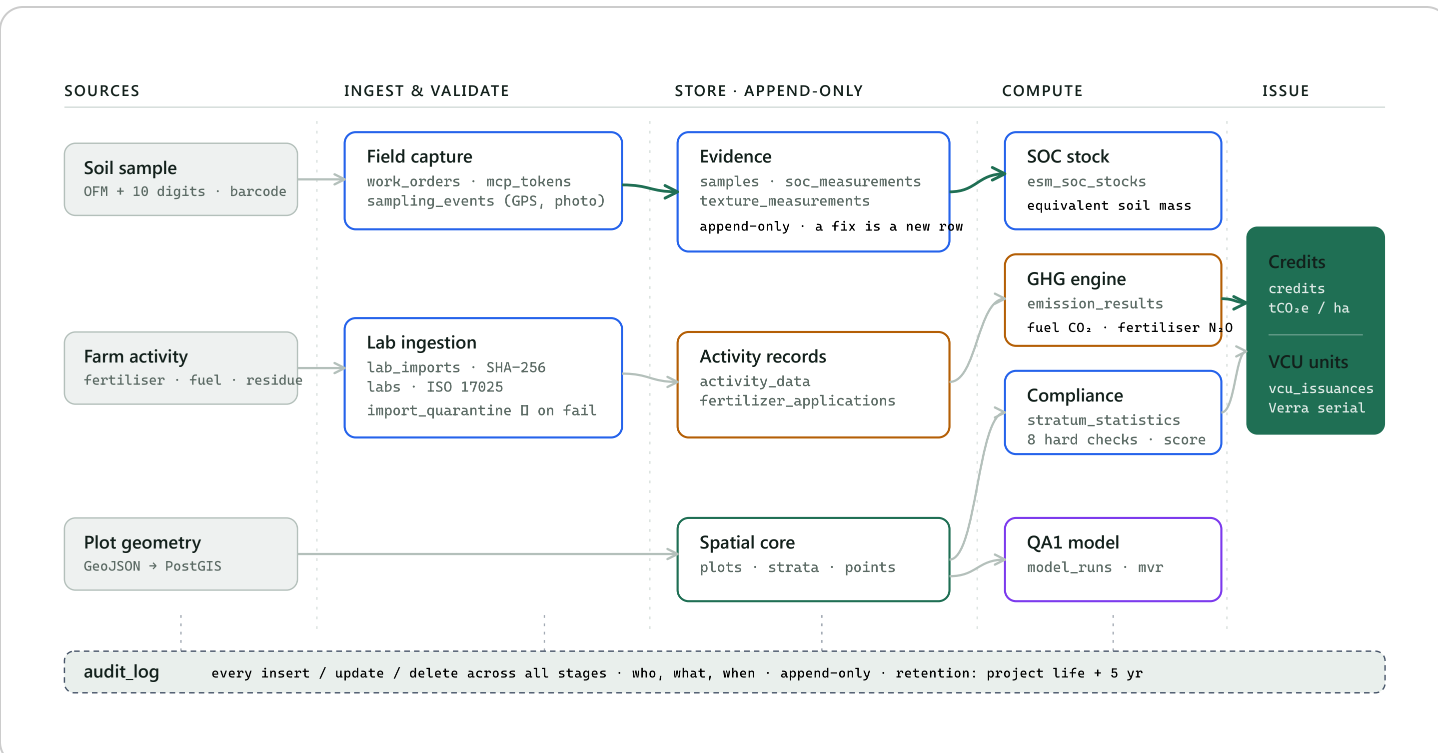

FIG 2 System Architecture

The seven schema domains as a layered stack, all inside one `mrv` Postgres schema on AWS. Two lineages — verification and commercial — meet only at the plot; a governance layer wraps everything; the AI layer sits ready for the module ahead.